Afro-Asian Forum for Innovation and Technology
AFFIAT · Aswan Finals 2026

PYGAME ZERO · RACING GAME

TURBO

Race smart. Drive safe.

Dodge traffic on six roads, collect coins for new cars, and learn a real road sign every time you crash.

01 · WHY Road rules are boring. Games are not.

THE PROBLEM المشكلة

Road signs feel like homework

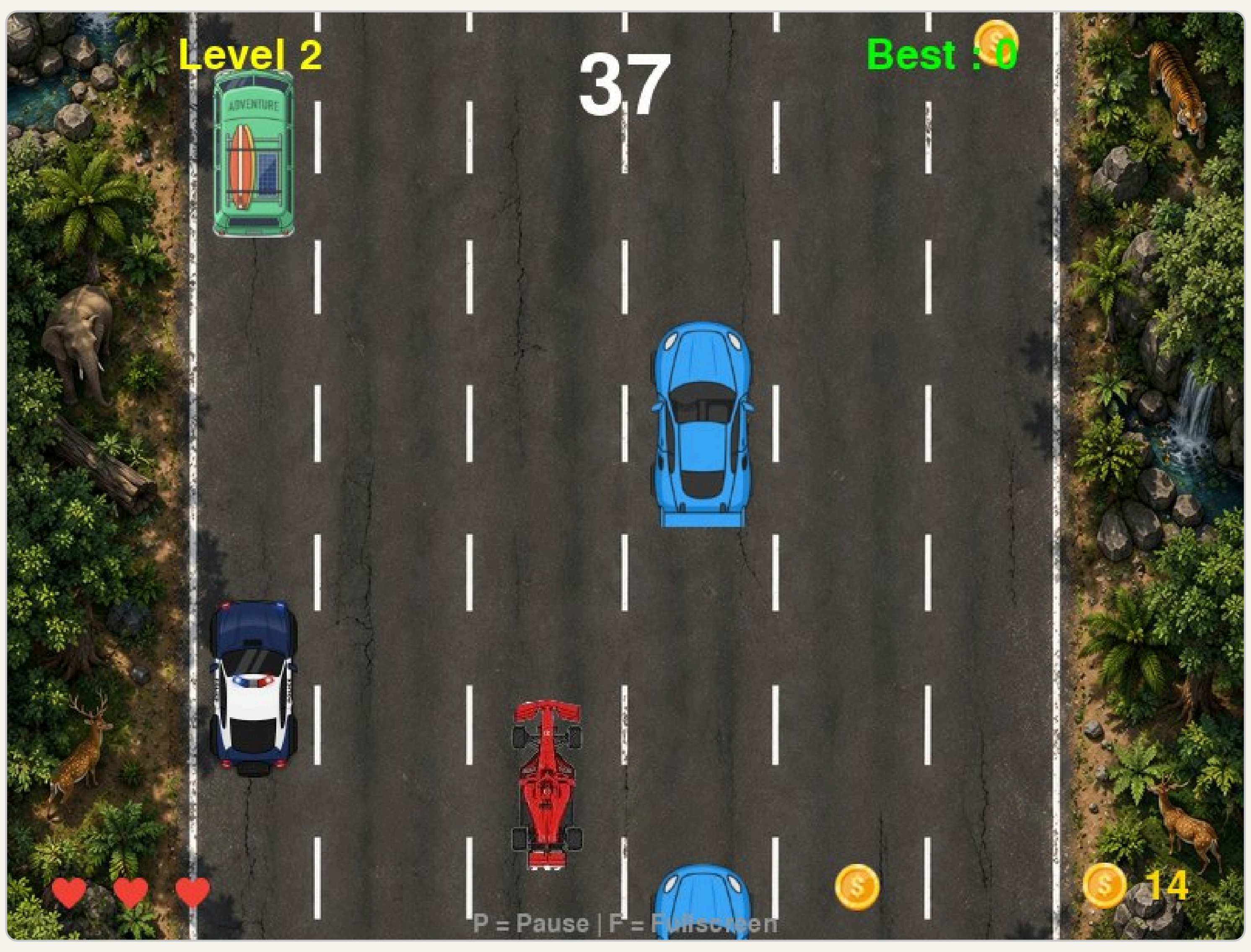
Children walk, cycle and ride in cars on busy roads every day. Road signs are taught as a long list to memorise, and most racing games reward speed and crashing, not safe driving.

OUR SOLUTION الحل

Learn a sign every race

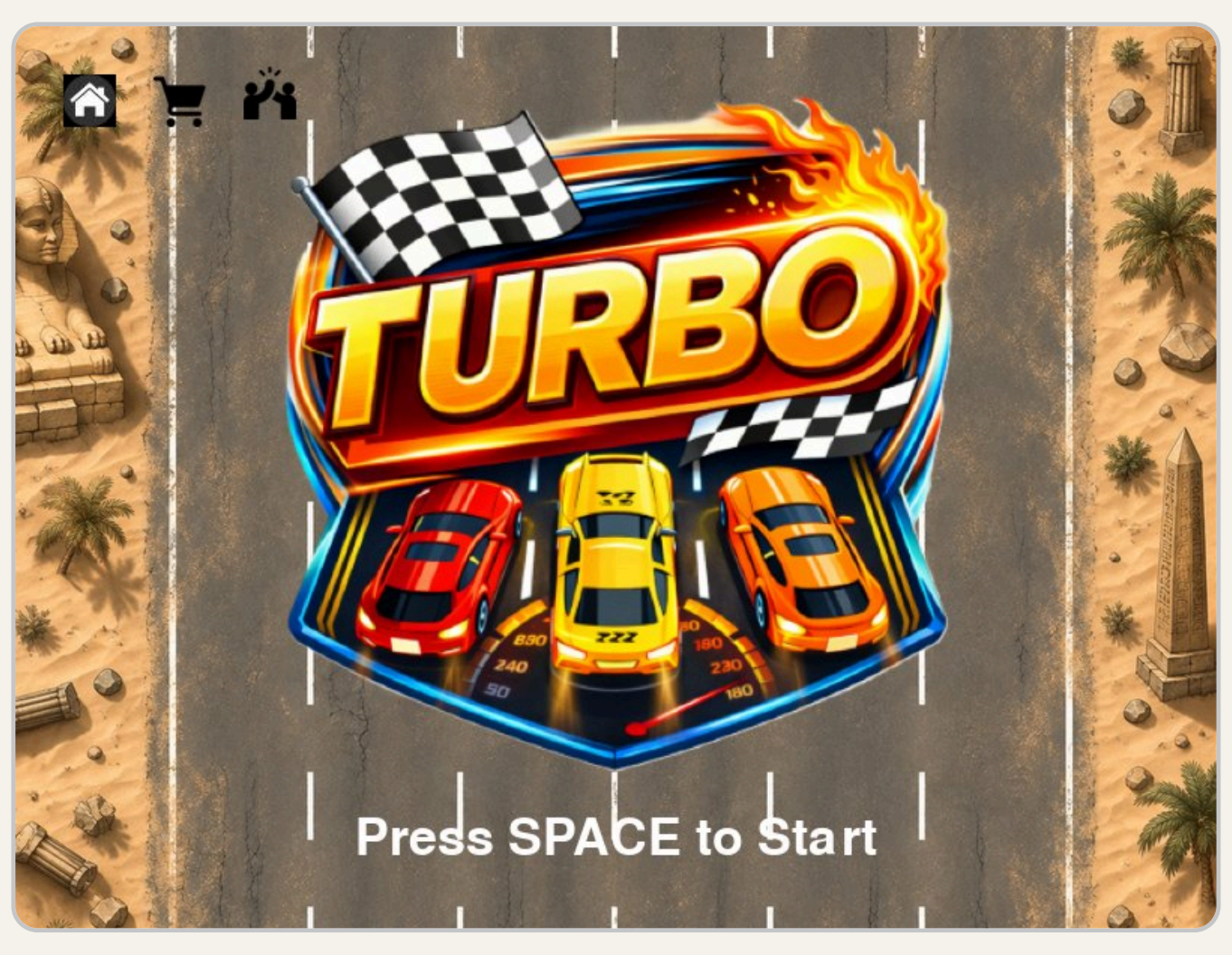
TURBO keeps all the fun of a racing game, then turns every crash into a short lesson: a real road sign and the rule behind it. Coins and new cars make you want one more race, and one more sign.

02 · SEE IT Dodge. Collect. Upgrade. تفادَ، اجمع، طوّر

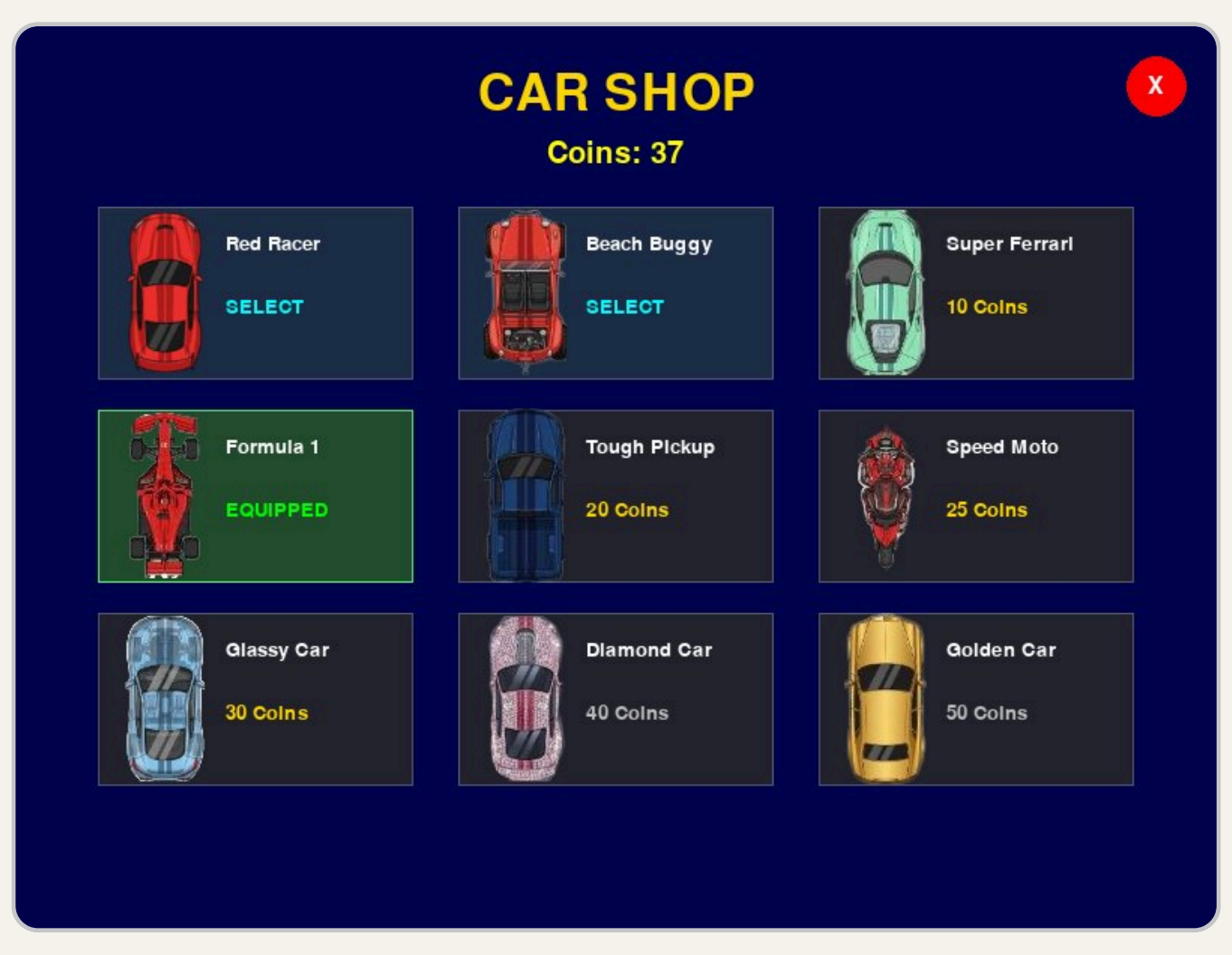


Level 2, the forest road: steer the Formula 1 car between the traffic and grab the coins. Score in the middle, hearts and coins at the bottom.

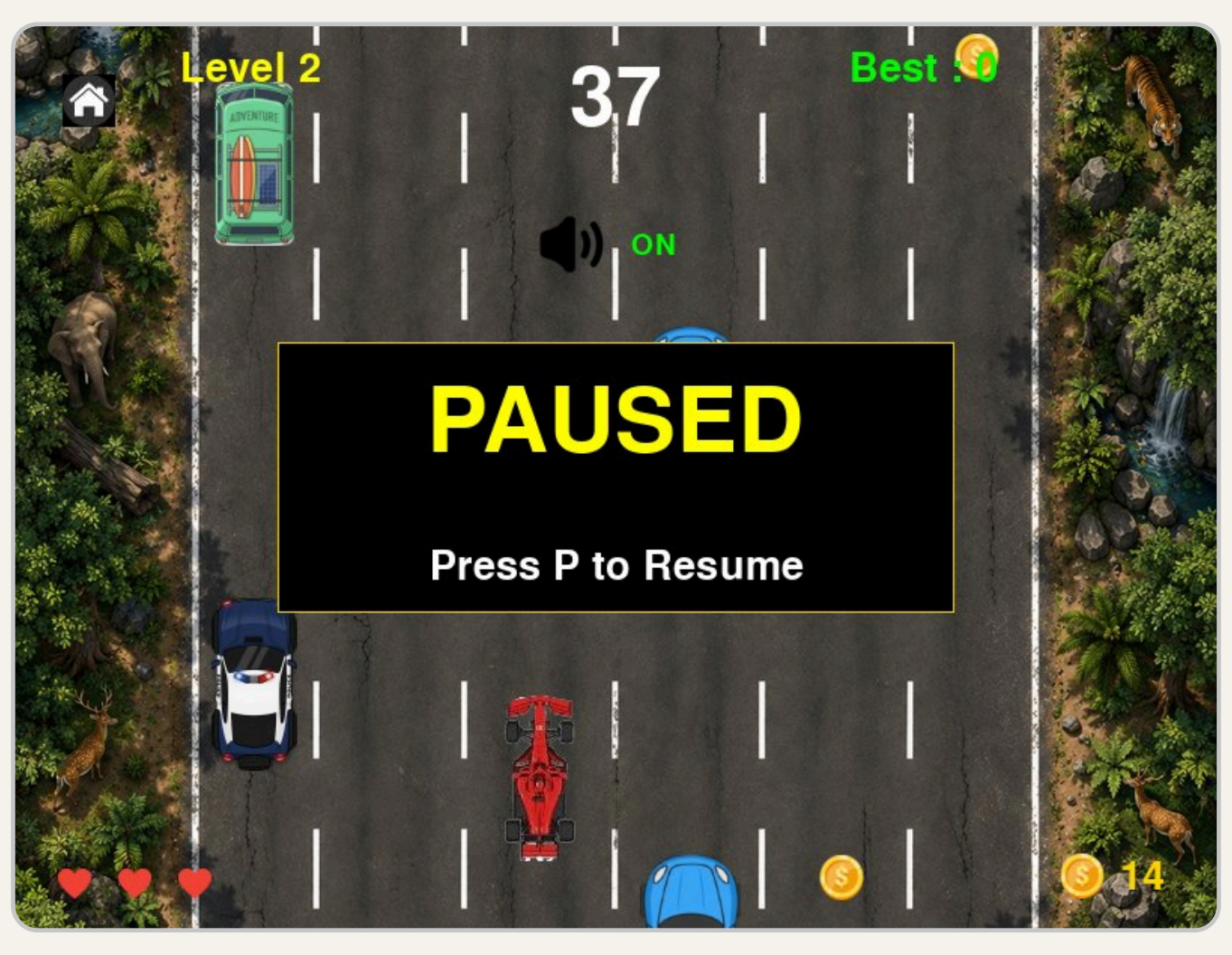
- **Dodge the traffic**
  Steer with ← → across 6 lanes. A crash costs 1 of your 3 hearts.
- **Six roads**
  Egypt, forest, snow, night, green fields and desert. A new road every 30 points.
- **It gets harder**
  Every 10 points the road speeds up and the cars come sooner.
- **Collect coins**
  A coin rolls in every 2.3 seconds. Touch it to hear the ding.
- **A car shop**
  Spend coins on 9 cars, from the free Red Racer to the 50-coin Golden Car.
- **Learn a road sign**
  Every game over shows 1 of 20 real road signs and what it means.



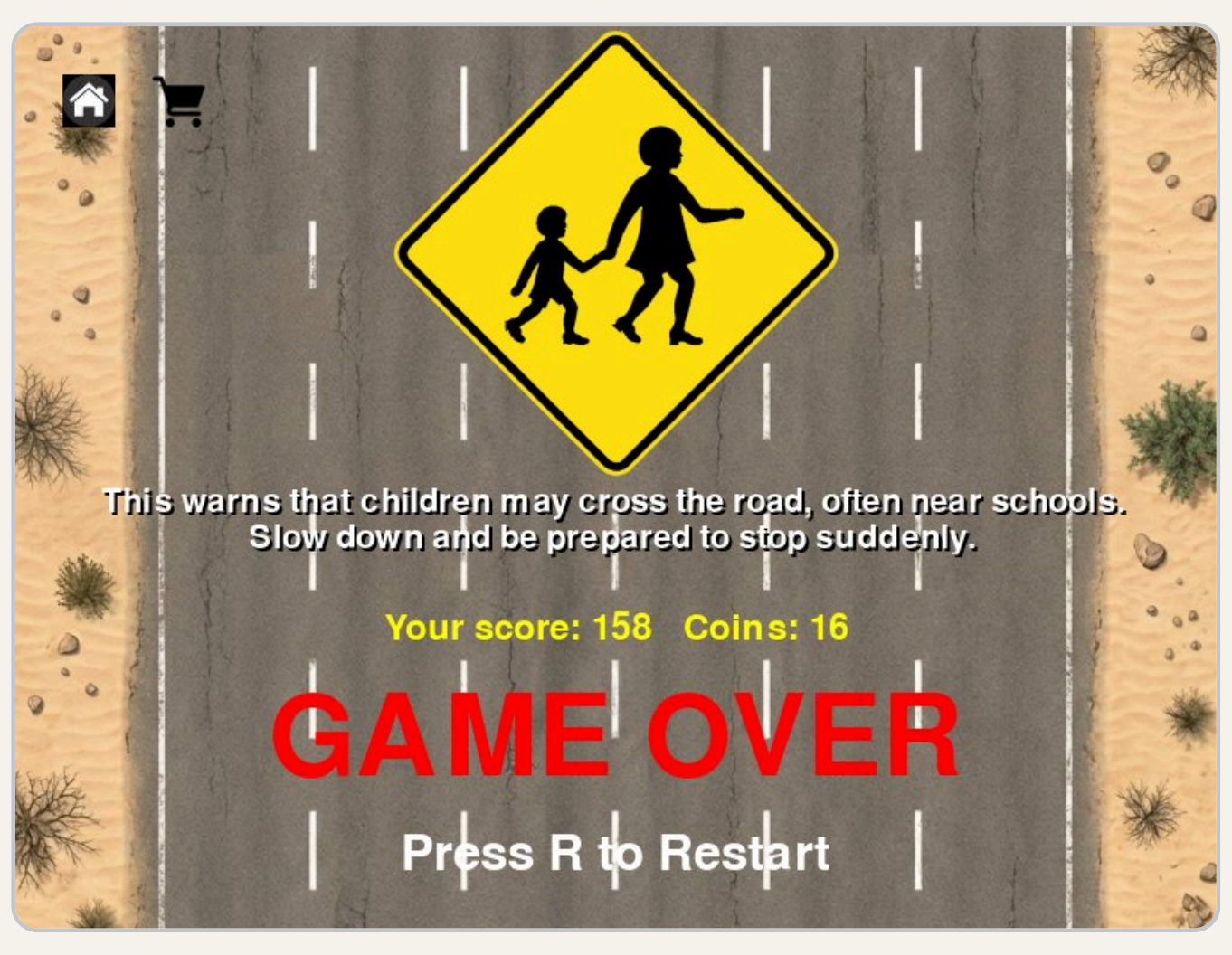
Start: the logo, and buttons for home, shop and makers.



Car shop: buy a car once, then pick it any time.



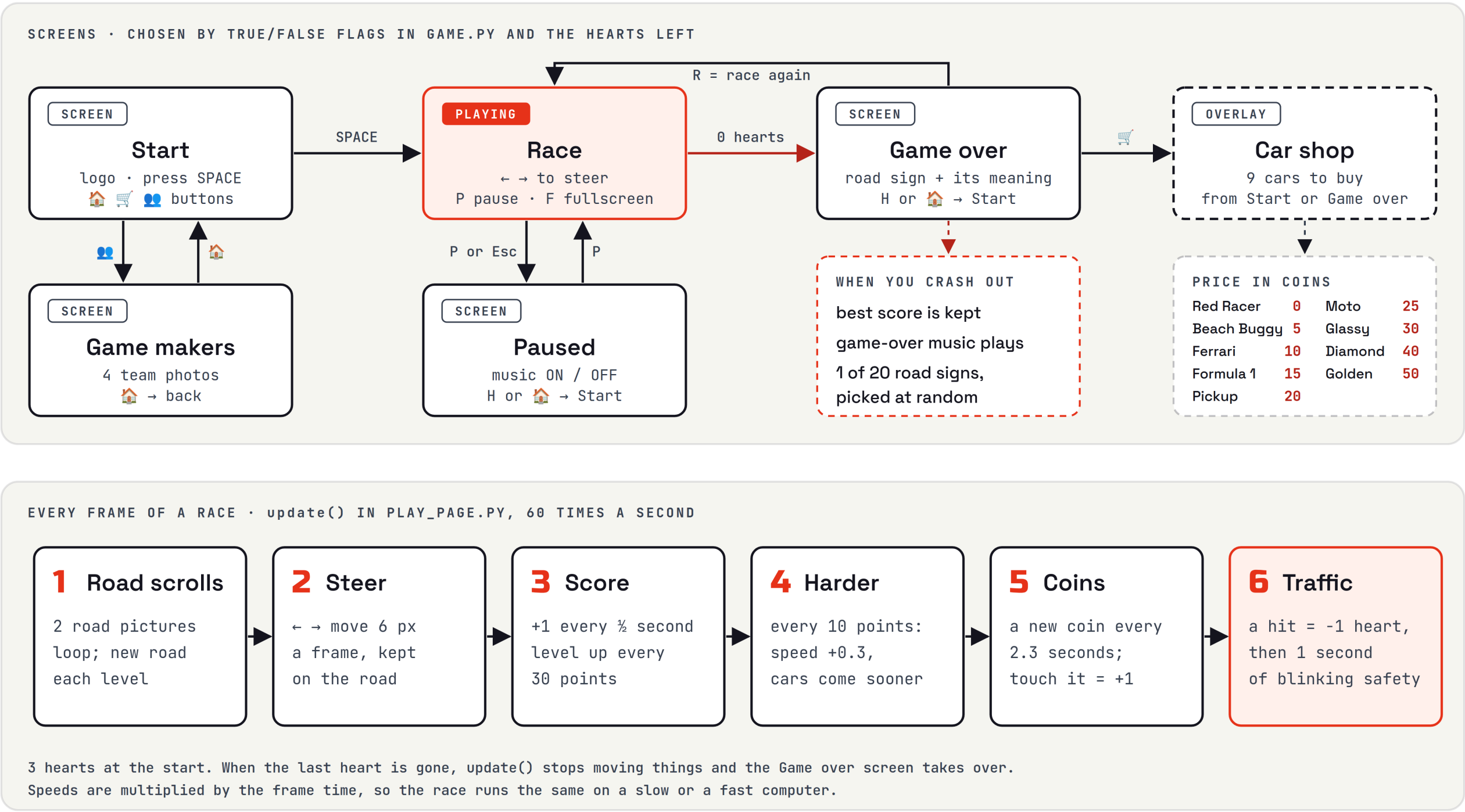
Paused: P to go on, and music on or off.


This warns that children may cross the road, often near schools. Slow down and be prepared to stop suddenly.

Game over: a road sign to learn, your score, R to race again.

03 · HOW Six roads, one game loop. ست طرق وحلقة لعب واحدة

SCREENS, AND ONE FRAME OF A RACE


SCREENS · CHOSEN BY TRUE/FALSE FLAGS IN GAME.PY AND THE HEARTS LEFT

WHEN YOU CRASH OUT
best score is kept
game-over music plays
1 of 20 road signs, picked at random

PRICE IN COINS

Red Racer	0	Moto	25
Beach Buggy	5	Glassy	30
Ferrari	10	Diamond	40
Formula 1	15	Golden	50
Pickup	20		

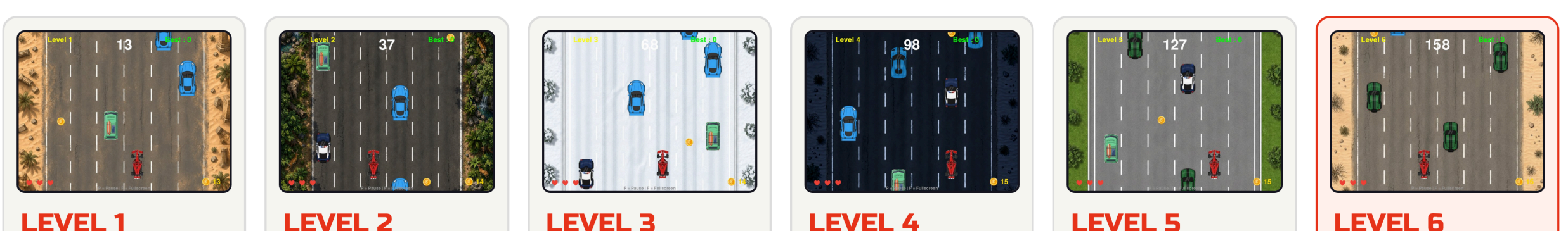
EVERY FRAME OF A RACE · update() IN PLAY_PAGE.PY, 60 TIMES A SECOND

1 **Road scrolls** 2 road pictures loop; new road each level →
2 **Steer** ← → move 6 px a frame, kept on the road →
3 **Score** +1 every ½ second level up every 30 points →
4 **Harder** every 10 points: speed +0.3, cars come sooner →
5 **Coins** a new coin every 2.3 seconds; touch it = +1 →
6 **Traffic** a hit = -1 heart, then 1 second of blinking safety

3 hearts at the start. When the last heart is gone, update() stops moving things and the Game over screen takes over. Speeds are multiplied by the frame time, so the race runs the same on a slow or a fast computer.

THE SIX LEVELS

SIX ROADS · A NEW ONE EVERY 30 POINTS (ABOUT 15 SECONDS)



	LEVEL 1 Egypt	LEVEL 2 Forest	LEVEL 3 Snow	LEVEL 4 Night	LEVEL 5 Green fields	LEVEL 6 Desert
points	0-29	30-59	60-89	90-119	120-149	150+
reached at	0:00	0:15	0:30	0:45	1:00	1:15
start speed	3.0	3.9	4.8	5.7	6.6	7.5 → 15
new car every	2.0 s	1.5 s	1.0 s	0.67 s	0.67 s	0.67 s

Faster every 10 points
The road, coins and cars speed up by 0.3. Top speed is 15, at 400 points.

More traffic
A new car every 2 seconds at the start. From 80 points: every 0.67 seconds, in one of 6 lanes.

Smooth road change
The old road fades out over 0.75 seconds while the new one scrolls in.

1
library to install: Pygame Zero

7
Python files: the main loop and one per screen

599
lines of code in all

0
classes: plain variables, ifs and loops

BUILT WITH

`Python` `Pygame Zero` `pygame`

Jomana & Ali · Team NINJA.PY

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Coins, cars and best score are not saved: closing the game starts fresh.

Scan for the code
github.com/mr-basem/Turpo